In [7]:
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#Hemant Marotkar , UID:- 23015002
import pandas as pd
from sklearn.preprocessing import MinMaxScaler,LabelEncoder
#load dataset
df=pd.read_csv("network_security_dataset.csv")

print("Original Dataset:")
print(df)

# remove duplicates rows
df=df.drop_duplicates()

#check missing values
df=df.drop_duplicates()

#check missing values
print("\nMissing values before cleaning:")
print(df.isnull().sum())

#fill missing numeric values with mean
numeric_cols=["src_bytes","dst_bytes","duration","packet_count","login_attempts"]

for col in numeric_cols:
    df[col]=df[col].fillna(df[col].mean())

#fill missing categorical values with mode
categorical_cols=["protocol","service","flag"]

for col in categorical_cols:
    df[col]=df[col].fillna(df[col].mode()[0])

#Encode categorical feature
label_encoder=LabelEncoder()

for col in categorical_cols:
    df[col]=label_encoder.fit_transform(df[col])

#encode target label
df["label"]=label_encoder.fit_transform(df["label"])

#normalize numeric feature
scaler=MinMaxScaler()
df[numeric_cols]=scaler.fit_transform(df[numeric_cols])

print("\nCleaned Dataset:")
print(df)
```

```
Original Dataset:
   connection_id protocol   service  src_bytes  dst_bytes  duration  \
0           C001      tcp      http      181.0     5450.0       0.5
1           C002      tcp       ftp      239.0      486.0       1.2
2           C003      udp       dns      146.0        0.0       0.1
3           C004      tcp      http        NaN     1337.0       2.3
4           C005     icmp     eco_i     1032.0        0.0       0.0
5           C006      tcp       ssh      321.0     2456.0       4.5
6           C007      tcp      http        0.0        0.0       0.0
7           C008      udp       dns       98.0      120.0       0.2
8           C009      tcp      smtp      250.0     3000.0       1.8
9           C010      tcp       ftp      400.0      100.0       3.0
10          C011      udp  domain_u       75.0        NaN       0.1
11          C012      tcp      http      181.0     5450.0       0.5
12          C013      tcp       ssh     1200.0      300.0      10.2
13          C014     icmp     eco_i      900.0        0.0       0.0
14          C015      tcp      http      210.0     1600.0       NaN
15          C016      udp       dns       85.0       90.0       0.1
16          C017      tcp      smtp      300.0     2500.0       2.0
17          C018      tcp       ssh     1500.0      200.0      12.0
18          C019      tcp      http      195.0     1400.0       0.7
19          C020      udp       dns       90.0       95.0       0.1

    packet_count  login_attempts  flag   label
0           24.0             1.0    SF  normal
1           18.0             3.0    SF  normal
2            4.0             0.0    SF  normal
3           50.0             2.0    SF  normal
4           10.0             0.0    SF  attack
5           70.0             NaN   REJ  attack
6            1.0             5.0  RSTO  attack
7            6.0             0.0    SF  normal
8           22.0             1.0    SF  normal
9           35.0             7.0   REJ  attack
10           5.0             0.0    SF  normal
11          24.0             1.0    SF  normal
12         130.0            10.0   REJ  attack
13           9.0             0.0    SF  attack
14          28.0             1.0    SF  normal
15           NaN             0.0    SF  normal
16          30.0             2.0    SF  normal
17         160.0            12.0  RSTO  attack
18          26.0             1.0    SF  normal
19           5.0             0.0    SF  normal

Missing values before cleaning:
connection_id     0
protocol          0
service           0
src_bytes         1
dst_bytes         1
duration          1
packet_count      1
login_attempts    1
flag              0
label             0
```

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dtype: int64

Cleaned Dataset:
   connection_id  protocol  service  src_bytes  dst_bytes  duration  \
0           C001         1        4   0.120667   1.000000  0.041667
1           C002         1        3   0.159333   0.089174  0.100000
2           C003         2        0   0.097333   0.000000  0.008333
3           C004         1        4   0.259754   0.245321  0.191667
4           C005         0        2   0.688000   0.000000  0.000000
5           C006         1        6   0.214000   0.450642  0.375000
6           C007         1        4   0.000000   0.000000  0.000000
7           C008         2        0   0.065333   0.022018  0.016667
8           C009         1        5   0.166667   0.550459  0.150000
9           C010         1        3   0.266667   0.018349  0.250000
10          C011         2        1   0.050000   0.237412  0.008333
11          C012         1        4   0.120667   1.000000  0.041667
12          C013         1        6   0.800000   0.055046  0.850000
13          C014         0        2   0.600000   0.000000  0.000000
14          C015         1        4   0.140000   0.293578  0.172368
15          C016         2        0   0.056667   0.016514  0.008333
16          C017         1        5   0.200000   0.458716  0.166667
17          C018         1        6   1.000000   0.036697  1.000000
18          C019         1        4   0.130000   0.256881  0.058333
19          C020         2        0   0.060000   0.017431  0.008333

    packet_count  login_attempts  flag  label
0       0.144654        0.083333     2      1
1       0.106918        0.250000     2      1
2       0.018868        0.000000     2      1
3       0.308176        0.166667     2      1
4       0.056604        0.000000     2      0
5       0.433962        0.201754     0      0
6       0.000000        0.416667     1      0
7       0.031447        0.000000     2      1
8       0.132075        0.083333     2      1
9       0.213836        0.583333     0      0
10      0.025157        0.000000     2      1
11      0.144654        0.083333     2      1
12      0.811321        0.833333     0      0
13      0.050314        0.000000     2      0
14      0.169811        0.083333     2      1
15      0.211188        0.000000     2      1
16      0.182390        0.166667     2      1
17      1.000000        1.000000     1      0
18      0.157233        0.083333     2      1
19      0.025157        0.000000     2      1
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